

Spring 1992

Problem 1 (Su77, Sp92). 1. Prove that every finitely generated subgroup of $\mathbb{Q}$, the additive group of rational numbers, is cyclic.

2. Does the same conclusion hold for finitely generated subgroups of $\mathbb{Q}/\mathbb{Z}$, where $\mathbb{Z}$ is the group of integers?

Note: See also ??????.

Problem 2 (Sp92). Find a square root of the matrix

$$\begin{pmatrix} 1 & 3 & -3 \\ 0 & 4 & 5 \\ 0 & 0 & 9 \end{pmatrix}$$

How many square roots does this matrix have?

Problem 3 (Sp92). Let f be an analytic function in the connected open subset G of the complex plane. Assume that for each point z in G , there is a positive integer n such that the n -th derivative of f vanishes at z . Prove that f is a polynomial.

Problem 4 (Sp92, Fa99). Show that every infinite closed subset of $\mathbb{R}^n$ is the closure of a countable set.

Problem 5 (Sp92). Let f be a differentiable function from $\mathbb{R}^n$ to $\mathbb{R}^n$. Assume that there is a differentiable function g from $\mathbb{R}^n$ to $\mathbb{R}$ having no critical points such that $g \circ f$ vanishes identically. Prove that the Jacobian determinant of f vanishes identically.

Problem 6 (Sp92). Let p be a prime integer, $p \equiv 3 \pmod{4}$, and let $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. If $x^4 + 1$ factors into a product $g(x)h(x)$ of two quadratic polynomials in $\mathbb{F}_p[x]$, prove that $g(x)$ and $h(x)$ are both irreducible over $\mathbb{F}_p$.

Problem 7 (Sp92). Let $a_1, a_2, \dots, a_{10}$ be integers with $1 \leq a_i \leq 25$, for $1 \leq i \leq 10$. Prove that there exist integers $n_1, n_2, \dots, n_{10}$, not all zero, such that

$$\prod_{i=1}^{10} a_i^{n_i} = 1$$

Problem 8 (Fa82, Sp92, Fa16). Evaluate $\int_{-\infty}^{\infty} \frac{\sin^3 x}{x^3} dx$.

Problem 9 (Sp92). Let p be a nonconstant polynomial with real coefficients and only real zeros. Prove that for each real number r , the polynomial $p - rp'$ has only real zeros.

Problem 10 (Sp92). Let A denote the matrix

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

For which positive integers n is there a complex 4×4 matrix X such that $X^n = A$?

Problem 11 (Sp92). Find a Laurent series that converges in the annulus $1 < |z| < 2$ to a branch of the function $\log \left(\frac{z(2-z)}{1-z} \right)$.

Problem 12 (Sp92). Let A be a real symmetric $n \times n$ matrix with nonnegative entries. Prove that A has an eigenvector with nonnegative entries.

Problem 13 (Sp92). Let f be a one-to-one $\mathcal{C}^1$ map of $\mathbb{R}^3$ into $\mathbb{R}^3$, and let J denote its Jacobian determinant. Prove that if x_0 is any point of $\mathbb{R}^3$ and $Q_r(x_0)$ denotes the cube with center x_0 , side length r , and edges parallel to the coordinate axes, then

$$|J(x_0)| = \lim_{r \rightarrow 0} r^{-3} \text{vol}(f(Q_r(x_0))) \leq \limsup_{x \rightarrow x_0} \frac{\|f(x) - f(x_0)\|^3}{\|x - x_0\|^3}$$

Problem 14 (Su77, Fa81, Sp92, Fa08). Prove that $\alpha = \sqrt{5} + \sqrt{7}$ is algebraic over $\mathbb{Q}$ by writing down a polynomial $f(x) \in \mathbb{Q}[x]$ of degree 4 having α as a zero. Show that $f(x)$ is irreducible over $\mathbb{Q}$.

Problem 15 (Sp92). Let $G \subset S_{999}$ be an abelian subgroup of order 1111. Prove that there exists $i \in \{1, \dots, 999\}$ such that for all $\sigma \in G$, one has $\sigma(i) = i$.

Problem 16 (Fa87, Sp92, Fa97, Sp00, Fa13, Sp19). Define a sequence of real numbers (x_n) by

$$x_0 = 1 \quad x_{n+1} = \frac{1}{2 + x_n} \quad \text{for } n \geq 0$$

Show that (x_n) converges and evaluate its limit.

Problem 17 (Sp92). For which positive numbers a and b , with $a > 1$, does the equation $\log_a x = x^b$ have a positive solution for x ?

Problem 18 (Sp92). Let the function f be analytic in the entire complex plane, real valued on the real axis, and of positive imaginary part in the upper half-plane. Prove $f'(x) > 0$ for x real.